

Probability in Modern Scanning Systems

Barcodes and QR Codes

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Introduction:

Barcodes and QR codes are essential tools in modern automated systems such as retail, logistics, manufacturing, and digital payments.

They rely on probability theory to reduce scanning errors and improve reliability.

Problem Statement:

Manual data entry is slow and error-prone. Probabilistic error models help prevent data loss in noisy environments.

What Are Barcodes and QR Codes:

Barcodes are one-dimensional codes using black and white stripes.

QR codes are two-dimensional matrix codes with higher capacity and error correction.

Probability Models:

Each bit has a probability p of being incorrect.

For n bits:

$$P(\text{error}) = 1 - (1 - p)^n$$

Redundancy:

Adding redundant bits increases the probability of successful decoding.

Applications:

Retail, logistics, payments, tracking systems.

Conclusion:

Probability theory ensures reliable data transmission under uncertainty.